

NSDA Level-4 Python with Django Advanced MCQ

Part-1: Django Model, ORM, Forms, Serializer + Python Advanced

1. What type of object does `range()` return in Python?

- A. List
- B. Tuple
- C. Range Object
- D. Dictionary

Ans: C

2. Which loop is guaranteed to execute at least once?

- A. for loop
- B. while loop
- C. do-while loop
- D. None in Python

Ans: D

3. What is the correct syntax of the Python Ternary Operator?

- A. if condition ? a : b
- B. a if condition else b
- C. if condition(a,b)
- D. condition(a:b)

Ans: B

4. What is the purpose of `elif`?

- A. To create loops
- B. To check multiple conditions
- C. To create functions
- D. To create classes

Ans: B

5. What does Nested Loop mean?

- A. A loop inside another loop
- B. A loop inside a function
- C. A function inside a class
- D. Stopping a loop

Ans: A

6. What does the `break` statement do?

- A. Skips the current iteration
- B. Completely stops the loop
- C. Stops the program
- D. Creates an error

Ans: B

7. What does the `continue` statement do?

- A. Stops the loop
- B. Skips the current iteration
- C. Returns from a function

D. Deletes a variable

Ans: B

8. What is a Default Argument in a Python Function?

A. An argument that has a default value

B. A required argument

C. A deleted argument

D. A global argument

Ans: A

9. What does `*args` accept?

A. Keyword arguments

B. Multiple positional arguments

C. Integer values

D. String values

Ans: B

10. What does `**kwargs` accept?

A. Multiple keyword arguments

B. List values

C. Tuple values

D. Integer values

Ans: A

11. Which architecture does Django follow?

A. MVC

- B. MVT
- C. MVP
- D. MVVM

Ans: B

12. What does the Django Model handle?

- A. Database structure
- B. User Interface
- C. URL routing
- D. CSS design

Ans: A

13. What is the responsibility of a Django View?

- A. Handle business logic
- B. Create databases
- C. Write CSS
- D. Install servers

Ans: A

14. Which layer is Django Template?

- A. Presentation Layer
- B. Database Layer
- C. Security Layer
- D. Server Layer

Ans: A

15. In which file are Django Models written?

- A. views.py
- B. urls.py
- C. models.py
- D. forms.py

Ans: C

16. Which class is used for Django Model Fields?

- A. models
- B. views
- C. forms
- D. admin

Ans: A

17. What is a characteristic of a Primary Key?

- A. It can have duplicates
- B. It must be unique
- C. It must be empty
- D. It must be a string

Ans: B

18. Which Django Field creates an auto-increment primary key?

- A. AutoField
- B. IntegerField
- C. CharField

D. BooleanField

Ans: A

19. Which Field is used to store long text data?

- A. TextField
- B. IntegerField
- C. DateField
- D. FloatField

Ans: A

20. Which Field is suitable for storing email addresses?

- A. EmailField
- B. TextField
- C. CharField
- D. URLField

Ans: A

21. What type of relationship does a ForeignKey create?

- A. One-to-One
- B. One-to-Many
- C. Many-to-Many
- D. None

Ans: B

22. What type of relationship does ManyToManyField create?

- A. One-to-One
- B. One-to-Many
- C. Many-to-Many
- D. One-to-None

Ans: C

23. What type of relationship does OneToOneField create?

- A. One-to-One
- B. One-to-Many
- C. Many-to-Many
- D. Many-to-One

Ans: A

24. Which command creates migrations in Django?

- A.
`python manage.py migrate`
- B.
`python manage.py makemigrations`
- C.
`python manage.py create`
- D.
`python manage.py model`

Ans: B

25. Which command applies migrations to the database?

- A. migrate
- B. makemigrations
- C. install
- D. update

Ans: A

26. What is the full form of Django ORM?

- A. Object Relational Mapping
- B. Object Record Manager
- C. Online Relation Model
- D. Object Runtime Method

Ans: A

27. Which query retrieves all objects from a Django model?

- A.
`Model.objects.all()`
- B.
`Model.objects.get()`
- C.
`Model.objects.first()`
- D.
`Model.objects.create()`

Ans: A

28. Which method is used to retrieve a single object?

- A. filter()
- B. get()
- C. all()
- D. values()

Ans: B

29. What does the `filter()` method return?

- A. Single Object
- B. QuerySet
- C. String
- D. Integer

Ans: B

30. What does the `exists()` method return?

- A. Object
- B. Boolean
- C. QuerySet
- D. List

Ans: B

31. Django Forms come from which package?

- A. django.forms
- B. django.models
- C. django.views
- D. django.template

Ans: A

32. Which one is used to create a form based on a Model?

- A. Form
- B. ModelForm
- C. ViewForm
- D. DatabaseForm

Ans: B

33. Which method is used for Form Validation?

- A. validate()
- B. clean()
- C. check()
- D. verify()

Ans: B

34. Which method is used for specific field validation?

- A. clean_fieldname()
- B. validate_field()
- C. field_clean()
- D. check_field()

Ans: A

35. Which method checks whether Django Form data is valid?

- A. is_valid()
- B. check()

C. validate()

D. verify()

Ans: A

36. What does a Django REST Framework Serializer do?

A. Converts objects into JSON

B. Creates databases

C. Runs servers

D. Creates HTML

Ans: A

37. Which Serializer is based on Django Models?

A. Serializer

B. ModelSerializer

C. JSONSerializer

D. ObjectSerializer

Ans: B

38. What is the full form of DRF?

A. Django Request Framework

B. Django REST Framework

C. Database REST Framework

D. Django Routing Framework

Ans: B

39. Which format is commonly used for API responses?

- A. XML Only
- B. JSON
- C. TXT
- D. CSV Only

Ans: B

40. Which method is used for Serializer validation?

- A. validate()
- B. check()
- C. verify()
- D. clean()

Ans: A

41. Which method is used to register a Model in Django Admin?

- A. admin.register()
- B. admin.site.register()
- C. register.admin()
- D. site.admin()

Ans: B

42. What is the Django Template variable syntax?

- A.
`{% variable %}`
- B.
`{{ variable }}`

C.

<variable>

D.

\$(variable)

Ans: B

43. Which tag is used for CSRF Protection in Django Forms?

A.

{% csrf_token %}

B.

{% security %}

C.

{% protect %}

D.

{% token %}

Ans: A

44. What is Django Middleware?

A. A request/response processing layer

B. A database

C. A template

D. A model

Ans: A

45. Which is the built-in Django Authentication app?

- A. django.auth
- B. django.contrib.auth
- C. django.authentication
- D. auth.django

Ans: B

46. Which function is used for user login in Django?

- A. login()
- B. signin()
- C. authenticate_user()
- D. user_login()

Ans: A

47. How does Django store passwords?

- A. Plain Text
- B. Using Hash Algorithm
- C. Encryption Only
- D. XML

Ans: B

48. Where can Django Session data be stored?

- A. Database
- B. Cache
- C. Cookie
- D. All of the above

Ans: D

49. Why is `select_related()` used in Django?

- A. Query Optimization
- B. Form Validation
- C. URL Routing
- D. Template Design

Ans: A

50. Why is `prefetch_related()` used in Django?

- A. To optimize related object queries
- B. To save passwords
- C. To create forms
- D. To create APIs

Ans: A

51. Which Python keyword is used to create an anonymous function?

- A. function
- B. lambda
- C. anonymous
- D. def

Ans: B

52. What is the output type of a function containing `yield`?

- A. List
- B. Generator

C. Tuple

D. Dictionary

Ans: B

53. Which operator checks whether two variables refer to the same object in memory?

A. ==

B. is

C. equals

D. ===

Ans: B

54. Which method is called automatically when an object is created?

A. start()

B. create()

C. **init()**

D. constructor()

Ans: C

55. What is the purpose of `super()` in Python?

A. To create a new object

B. To call parent class methods

C. To delete a class

D. To import modules

Ans: B

56. Which Python concept allows the same method name to behave differently?

- A. Encapsulation
- B. Polymorphism
- C. Inheritance
- D. Abstraction

Ans: B

57. Which block is used to handle exceptions in Python?

- A. if-else
- B. try-except
- C. switch-case
- D. catch-finally

Ans: B

58. Which keyword is used to manually raise an exception?

- A. throw
- B. error
- C. raise
- D. exception

Ans: C

59. What does the `finally` block do?

- A. Executes only when error occurs
- B. Executes only when no error occurs
- C. Always executes
- D. Stops execution

Ans: C

60. Which function returns the length of an object?

- A. size()
- B. count()
- C. len()
- D. length()

Ans: C

61. Django follows which architectural pattern?

- A. MVC
- B. MVT
- C. MVP
- D. MVVM

Ans: B

62. Which file contains Django application settings?

- A. views.py
- B. models.py
- C. settings.py
- D. urls.py

Ans: C

63. Which command creates a new Django project?

- A. django-admin startproject
- B. django createproject
- C. python createproject
- D. django new

Ans: A

64. Which command creates a new Django application?

- A. django-admin startapp
- B. python createapp
- C. django newapp
- D. manage.py createapp

Ans: A

65. Which Django component handles business logic?

- A. Template
- B. Model
- C. View
- D. URL

Ans: C

66. What is the main purpose of Django ORM?

- A. Design HTML pages
- B. Communicate with databases using Python objects
- C. Manage CSS files
- D. Create JavaScript code

Ans: B

67. Which method retrieves all records from a Django model?

- A. get()
- B. all()
- C. filter()
- D. fetch()

Ans: B

68. Which method returns a single object?

- A. filter()
- B. all()
- C. get()
- D. values()

Ans: C

69. What does Django `filter()` return?

- A. Single object
- B. QuerySet
- C. Dictionary
- D. String

Ans: B

70. Which QuerySet method is used to sort data?

- A. order_by()
- B. sort()
- C. arrange()
- D. ordering()

Ans: A

71. Which field creates a relationship between two models?

- A. CharField
- B. ForeignKey
- C. TextField
- D. IntegerField

Ans: B

72. ForeignKey represents which relationship?

- A. One-to-One
- B. One-to-Many
- C. Many-to-Many
- D. Many-to-One only

Ans: B

73. Which field is used for many-to-many relationships?

- A. ManyField
- B. ManyToManyField
- C. MultipleField
- D. RelationField

Ans: B

74. What is the purpose of Django migrations?

- A. Create HTML pages
- B. Apply database schema changes
- C. Start server
- D. Create users

Ans: B

75. Which command applies migrations to the database?

- A. migrate
- B. makemigrations
- C. install
- D. update

Ans: A

76. What does `__str__()` method do in Django models?

- A. Converts object into readable string representation
- B. Deletes object
- C. Saves object
- D. Creates database table

Ans: A

77. What is the purpose of **Meta class inside a Django model?**

- A. Configure model options
- B. Create views
- C. Handle forms
- D. Create URLs

Ans: A

78. Which attribute makes a field optional in forms?

- A. null=True
- B. blank=True
- C. optional=True
- D. empty=True

Ans: B

79. Which attribute allows NULL values in database?

- A. blank=True
- B. null=True
- C. empty=True
- D. nullable=True

Ans: B

80. Which Django class is used to create HTML forms?

- A. `django.forms.Form`
- B. `django.html.Form`
- C. `django.template.Form`
- D. `django.web.Form`

Ans: A

81. What is `ModelForm` used for?

- A. Creating forms from models
- B. Creating database tables
- C. Creating APIs
- D. Creating templates

Ans: A

82. Which method validates Django forms?

- A. `check()`
- B. `validate()`
- C. `is_valid()`
- D. `verify()`

Ans: C

83. Which method is used for custom form validation?

- A. `clean()`
- B. `validate_custom()`
- C. `check_form()`
- D. `verify()`

Ans: A

84. What is Django REST Framework mainly used for?

- A. Creating APIs
- B. Creating databases
- C. Designing templates
- D. Managing files

Ans: A

85. Which serializer is directly connected with Django models?

- A. JSONSerializer
- B. ModelSerializer
- C. HTMLSerializer
- D. DataSerializer

Ans: B

86. Serializer converts complex data into:

- A. Python objects only
- B. JSON format
- C. Database tables
- D. HTML pages

Ans: B

87. Which HTTP method is used to retrieve data?

- A. POST
- B. PUT
- C. GET
- D. DELETE

Ans: C

88. Which HTTP method is used to create new data?

- A. POST
- B. GET
- C. DELETE
- D. PATCH

Ans: A

89. Which HTTP method is commonly used for partial update?

- A. POST
- B. PATCH
- C. GET
- D. DELETE

Ans: B

90. Which HTTP method removes a resource?

- A. REMOVE
- B. DELETE
- C. DROP
- D. CLEAR

Ans: B

91. What is middleware in Django?

- A. Layer that processes requests and responses
- B. Database table
- C. Template file
- D. Model class

Ans: A

92. Which Django feature protects against Cross-Site Request Forgery?

- A. Password hashing
- B. CSRF Token
- C. Session
- D. Cache

Ans: B

93. Which Django module provides authentication?

- A. django.auth
- B. django.contrib.auth
- C. django.authentication
- D. django.user

Ans: B

94. What is a Django signal?

- A. Event notification mechanism
- B. URL pattern
- C. Database query
- D. HTML component

Ans: A

95. When does `post_save` signal execute?

- A. Before saving object
- B. After saving object
- C. Before deleting object
- D. During migration

Ans: B

96. What is the purpose of `select_related()`?

- A. Optimize database queries using JOIN
- B. Create forms
- C. Validate data
- D. Create serializers

Ans: A

97. What is the purpose of `prefetch_related()`?

- A. Optimize many-to-many and reverse relationships
- B. Delete cache
- C. Create models
- D. Validate forms

Ans: A

98. Which Django file defines URL routing?

- A. `views.py`
- B. `urls.py`
- C. `models.py`
- D. `forms.py`

Ans: B

99. Which command creates a Django superuser?

- A. `createadmin`
- B. `createsuperuser`
- C. `admincreate`
- D. `makeadmin`

Ans: B

100. What is the default database in a new Django project?

A. MySQL

B. PostgreSQL

C. SQLite

D. Oracle

Ans: C